



Artificial intelligence applications in fake review detection: Bibliometric analysis and future avenues for research

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ABSTRACT

Artificial intelligence (AI) and machine learning (ML) are emergent tools for fake review detection. This study provides a comprehensive overview of AI applications in fake review detection, using a hybrid integrated review and by combining bibliometric analysis with a framework-based review: the 4Ws (What, Where, Why, and How). We draw on the thematic structure of AI and ML research in fake review detection for the 2012–2021 period by conducting bibliometric coupling, keyword co-occurrence, and conceptual thematic, social network, and cluster-based content analyses of scientific articles. The findings indicate that field research has thus far concentrated on three overarching groups of fake review detection: (a) word of mouth, quality, reputation, and price; (b) classification, moderating role, intention, and analytics; and (c) impact and participation. This article provides researchers, companies and policymakers with insights into how AI can detect fraudulent reviews and drives further research on the adequate use of advanced AI techniques in fake review detection.

1. Introduction

Online product reviews significantly influence customers' purchasing behaviors. However, fake reviews can mislead consumers, thus compromising the fairness of the online shopping environment. In 2012, the UK Advertising Standards Authority found that TripAdvisor was involved in creating fake reviews: approximately 50 million online reviews on its site could not be verified as being trustworthy. In 2013, after being caught paying for fake reviews and negative comments about competitors, Samsung was fined in excess of US\$340,000 by Taiwan's Fair Trade Commission (Souppouris, (2013, October 24), 2013). In 2015, Amazon took legal action in the Seattle lawsuit against 1,114 individuals who claimed to have posted fake reviews on its website. This came after Amazon used certain websites in April 2015 known for selling fake reviews (Cardy, (2015, October 18), 2015). In November 2019, the European Parliament and the Council of the European Union adopted Directive 2019/2161 regarding the better enforcement and modernization of union-wide consumer protection rules (Directive on the Modernization of Consumer Protection). This directive more closely aligned consumer protection with the digital era, ensured improved enforcement of those rules, and included additional rules regulating fake

reviews.

Although methods of detecting fake reviews have progressed, there is still a need to improve the accuracy of such detection by integrating work from psychology and computational linguistics. Thus, Ott et al. (2011) developed and compared three approaches for detecting fake opinion spams and ultimately compiled a classifier that was approximately 90 % accurate with regards our gold standard opinion spam dataset. Savage et al. (2015) used binomial regression to identify reviewers with an anomalous proportion of ratings deviating from the majority opinion. Wang et al. (2012) proposed the concept of a review graph to capture the relationships between all reviewers, reviews, and reviewed companies as a heterogeneous graph. They explored how interactions between nodes in this graph could reveal the cause of spam and proposed an iterative computation model to identify suspicious reviewers.

In recent years, researchers have moved beyond traditional methods to investigate machine learning (ML) and artificial intelligence (AI) (Cao et al., 2011; Fang et al., 2020; Merigó et al., 2015). For example, Ahmed et al. (2018) proposed an *n*-gram model to detect automatically fake content, particularly fake reviews and fake news. They studied and compared two different feature extraction techniques and six ML

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classification techniques. Wang et al. (2018) proposed a Latent Dirichlet Allocation (LDA)-based computing framework, namely, Graph-based Group SPAM (GGSpam), for group spam detection in product review data. Fang et al. (2020) considered that fake comments could be detected by incorporating comment time. To investigate this, they extracted six time points as distinguishing features from time-series data and designed two classification algorithms.

The application of AI, ML, and deep learning (DL) techniques to detect fake reviews has been the subject of several academic works and research in a constantly growing field. However, co-thematic research is still new and has not yet been fully explored. Thus, following Chakma et al. (2021), we used a hybrid integrated review combining bibliometric analysis with framework-based review. In our study, we used the 4Ws (What, Where, Why, and How) to provide a better structure for this review. According to Chakma et al. (2021), analytical synthesis of prior research helps to develop a topic or subject, assists academics with state-of-the-art perspectives on a specific research topic, and provides support

in identifying gaps in research. Thus, conducting a hybrid review provides a more robust understanding of fake review detection research, in addition to aiding the development of avenues for future research (Paul, Merchant, et al., 2021).

2. Review protocol, design and structure (SPAR-4-SLR)

The grounding of a protocol is essential for systematic literature reviews, as a protocol guarantees vigilant planning, uniformity in execution, and transparency enabling replication. There are a limited number of protocols for systematic reviews. Moher et al. (2009) and Moher et al. (2015) propose systematic review and meta-analysis (PRISMA) protocols (PRISMA-P). Thus, those protocols presented little justification researchers can use to explain their review choices. Paul et al. (2021) suggest an alternative protocol to address these limitations: the Scientific Procedures and Rationales for Systematic Literature Reviews protocol (SPAR-4-SLR protocol). We shape our literature reviews based on

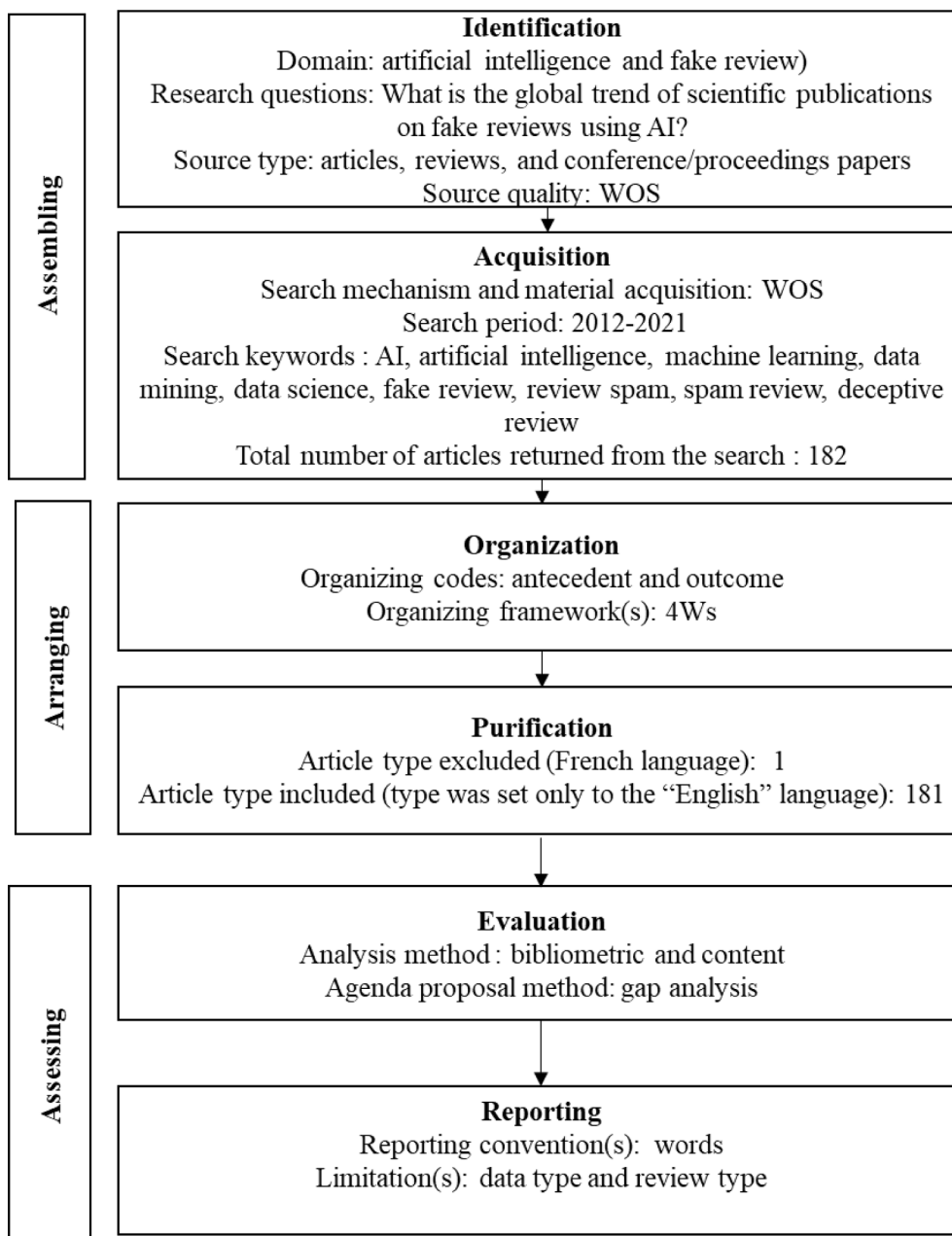


Fig. 1. Application of the SPAR-4-SLR protocol.

the SPAR-4-SLR protocol: assembling, arranging, and assessing literature. As shown in Fig. 1, this model comprises three stages and six substages.

We began the assembling stage by identifying the domain: Artificial intelligence applications in fake review detection. We then identified our research questions: What global trend can be observed in scientific publications on fake reviews using AI? What information can be revealed from this trend? Thirdly, we identified the source type (articles, reviews, and conferences/proceedings papers) and the source quality (WOS).

We began the Acquisition substage using Web of Science (WoS). WoS is regarded as the most accurate literature-indexing tool for technical and scientific information, providing insights into the most vital fields of scientific and technological study. Scholars have concluded that WoS fulfills higher quality standards than Scopus (Forliano et al., 2021). According to Merigó et al. (2015), WoS produced fewer false-positive results in terms of authors and keyword disambiguation as a consequence of the inclusion of keywords plus. In our study, we conducted an assessment to summarize the research addressing the impacts of AI applications on fake review detection. The search for publications to include in the review began from January 2012 to November 10, 2021, and we devised and followed a three-stage search approach: database searches, academic filtering, and language filtering (see Table 1). We used a wide range of search keywords in accordance with previous reviews (e.g., Wu et al., 2020; Goodell et al., 2021) as follows: (“AI” OR “artificial intelligence” OR “machine learning” OR “data mining” OR “data science” OR “algorithm” OR “analytics” OR “robot” OR “automation” OR “big data” OR “text-mining” OR “natural language processing” OR “soft computing”) and (“fake review” OR “review spam” OR “spam review” OR “deceptive review” OR “review manipulation” OR “fraudulent review” OR “review fraud” OR “false review” OR “suspicious review” OR “fictitious review” OR “manipulative review” OR “authentic review” OR “fake online review”). A total of 182 results were initially obtained.

Then, we started the Arranging substage, choosing the organizing framework. There are numerous types of reviews: structured reviews based on theories, constructs, and methods (Kahiya, 2018; Paul & Singh, 2017) reviews based on frameworks (Goodell et al., 2021; Lim et al., 2021); hybrid/integrated reviews (Bahoo et al., 2020; Dabić et al., 2020); theory-focused reviews (Gilal et al., 2019); reviews based on meta-analysis (Knoll & Matthes, 2017); and bibliometric studies (Bal-louk et al., 2021; Ruggeri et al., 2019). To provide a comprehensive overview of AI applications in fake review detection, we used a hybrid integrated review in this study. In accordance with Chakma et al. (2021), we combined bibliometric analysis with framework-based

review, the 4Ws following Chakma et al. (2021). In the Purification substage, the document type was set only to the “English” language, and duplicates were also eliminated using CiteSpace.

As a consequence of database searches and academic and language filtering, the remaining 181 articles that passed the screening procedure were prepared for bibliometric assessment. Following recent bibliometric studies (e.g., Xu et al., 2021), our analysis included articles, reviews, and conference/proceedings papers; our choice was also justified by the number of studies published in the field of computer sciences. A large number of documents in our database were published in the form of conference or proceedings papers. According to Lisée et al. (2008), proceedings play a particularly important role in computer sciences, accounting for close to 20 % of all references. In the field of bibliometrics, authors such as Butler & Visser (2006) and Glänzel et al. (2006) have argued that using conference and proceedings databases produces a more complete and, thus, precise picture of the scientific production of a given discipline. Furthermore, Butler et al. (2008) claim that, in the transfer of knowledge, proceedings are given even more weight than papers in computer sciences and other new subjects.

In the Assessing substage. First, we used bibliometrics to facilitate the identification of the essential quantitative variables of a particular research stream (Junquera & Mitre, 2007). This method enables us to determine the necessary information about a specific research topic, including authors in the field, the number of publications, and keywords, thus enabling interaction between variables (AI and fake review) and data across different countries (Casadesus-Masanell & Ricart, 2011). This method also enables the implementation of science mapping techniques (Aria & Cuccurullo, 2017). In this study, we conducted bibliometric analyses using the bibliometrix R-package (Aria & Cuccurullo, 2017), and VOSviewer, an application providing a web interface for bibliometrix, was also used to create the topic, a conceptual map, and a trend topic figure. To identify the most active geographies for this research area, we used Bing, and for the most active cluster, we used CiteSpace software. We then used the framework-based review design following Rosado-Serrano et al. (2018) and applied the 4Ws to answer the following research question: “What do we know about AI applications in fake review detection?”.

The bibliometric analysis achieved the following objectives:

- (A) Provide bibliometric information on 181 scientific studies extracted from the WoS database.
- (B) Use the bibliometrix R-package and CiteSpace to obtain and record the quantitative data of different selected articles.
- (C) Use variables, such as authors per article and an author’s dominance index, to identify the leading authors in this research area.
- (D) Use citations analysis and collaboration map to understand the network of this research stream.

In Section 4.1 (“An overview”), we present a general picture of fake review detection in an attempt to answer the question: “Why do we need to know more about AI applications in fake review detection?”. In Section 4.2 (“The effect of fake reviews”), we present the impacts of fake reviews and the need to use AI to detect them. Databases and methodologies in prior research (where and what kinds of research have been conducted) were examined. In Section 5, we highlight the different databases and methods used in prior studies on the application of AI techniques to detect fake reviews and to critically assess whether the chosen fake review detection methods have facilitated sufficient fake review detection tools. Finally, in Section 6, we answer the question: “How can this review help to fill the gaps in the literature and provide guidelines for future research?”. The findings provide a proposed agenda to guide future researchers in addressing gaps in the literature, which we present in section 6.

- (e) Use hot research topics and timelines of co-citation clusters to identify trends and avenues for future research.

Table 1
Web of science search query.

Search query	Results
Search date:	10 November 2021
Search topic:	182
	(“AI” OR “artificial intelligence” OR “machine learning” OR “data mining” OR “data science” OR “algorithm” OR “analytics” OR “robot” OR “automation” OR “big data” OR “text mining” OR “natural language processing” OR “soft computing”) AND (“fake review” OR “review spam” OR “spam review” OR “deceptive review” OR “review manipulation” OR “fraudulent review” OR “review fraud” OR “false review” OR “suspicious review” OR “fictitious review” OR “manipulative review” OR “authentic review” OR “fake online review”)
Document type:	182
	Include only articles, reviews, and conference/proceedings paper
Language screening:	181
	Include only English language-published document
Data extraction and analysis:	181

3. Results of the bibliometric analysis

3.1. Data analysis

For the data analysis, the bibliometrix R-package, VOSviewer 1.6.15, and CiteSpace 5.3R4 were adopted for scientometric analysis and assessment of map cooperation networks (authors, journals, and countries), along with keyword co-occurrence, document co-citation clustering, and journal dual-map overlay. Visualization of these collaborative networks in a given discipline is based on the theory of co-citation, which posits that two documents share a co-citation correlation when cited together in another document. Co-citation analysis, as proposed by Small (1973), aids in mapping the intellectual structure of a discipline as well as the exploration of the intellectual linkages which exist between influential articles in a given field (Culnan et al., 1990; Hota et al., 2020). Usually, co-citation analysis is performed for mapping older papers, and it is best performed within different time periods (Aria and Cuccurullo, 2017).

3.1.1. Number of publications per year

The annual distribution of publications between 2012 and 2021 is shown in Fig. 2. As can be seen, the number of publications began to gradually increase between 2012 and 2017. A significant increase was observed after 2014, with a marked breakthrough in 2018, showing a significant upward trend. The foregoing notwithstanding, the number of publications on AI applications in fake review detection between 2017 and 2021 was stable.

3.1.2. Country/region analysis

To date, 22 countries have contributed to this research ($N = 324$). Fig. 3 shows the geographical distribution of the number of publications per country. As can be seen, the highest number of contributions in terms of deploying AI in the detection of fake reviews came from China (137), followed by the United States (62), India (45), Australia (10), and Canada, Italy, and Pakistan (7 each). As shown in Fig. 4, the nodes and links represent the contributing countries and their level of cooperation, respectively. Based on the sizes of the nodes and the thickness of the lines in Fig. 4, we can clearly determine the contributions of individual countries and how they relate to one another. As shown in Fig. 4, China occupied a central position in this area and frequently cooperated with

the United States, Japan, South Korea, and Singapore. India, Canada, Finland, Taiwan, Denmark, and Bangladesh were closely related to each other, as were Australia and Indonesia.

3.1.3. Author and co-authorship analysis

Fig. 5 shows the top 10 most productive authors. Notably, the total numbers of publications of the top one to five authors were the same (four articles), namely for Jena SK, Rout JK, Wang Z, Yoshida T, and Zhang W, respectively. Similarly, the authors ranked in fifth to tenth place published equivalent quantities of articles.

As shown in Fig. 6, most authors were most active between 2014 and 2019. Over time, the number of authors in the field increased, and connections among them became more frequent (Fig. 7). In total, 518 authors contributed to 181 publications on AI applications in fake review detection, with several groups of authors forming small circles of cooperation (Fig. 6). Notably, all eight of the most cited authors published their research in 2012 (Table 2). In addition, Fig. 6 and Table 2 reveal a relative lack of communication among the most cited authors.

3.1.4. Journal analysis

A journal analysis was conducted to assess the degree of influence of different publications. However, as the total number of articles published in journals varies, it may be insufficient to evaluate their influence using statistical data analysis alone. Thus, a co-citation analysis was used to visualize internal relationships among journals. Overall, 143 journals and conferences published 181 papers in this field, and 240 citations were published between them. As shown in Fig. 8, the most productive journal in this field was *Expert Systems with Applications* (7 articles), followed by *IEEE Access Journal* (6 articles). These are consistent with the results shown in Fig. 9. The sizes of the nodes indicate that *Expert Systems with Applications* was the most frequently cited journal, followed by *IEEE Access Journal*.

3.2. Hot research topics on fake reviews in the field of AI

The spread and progress of research topics reflect changes in focus areas, analytical perspectives, and research methods which took place at different times. Co-citation, as a refined expression of the research topic of academic papers, can reveal the internal relationships of knowledge

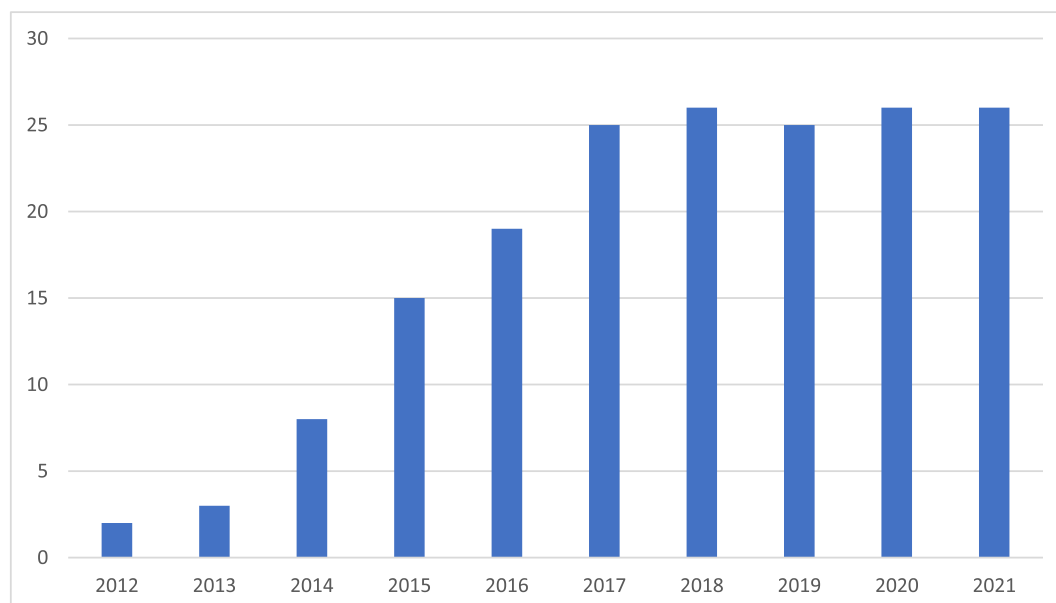


Fig. 2. Time evolution of total publications in the WOS databases.

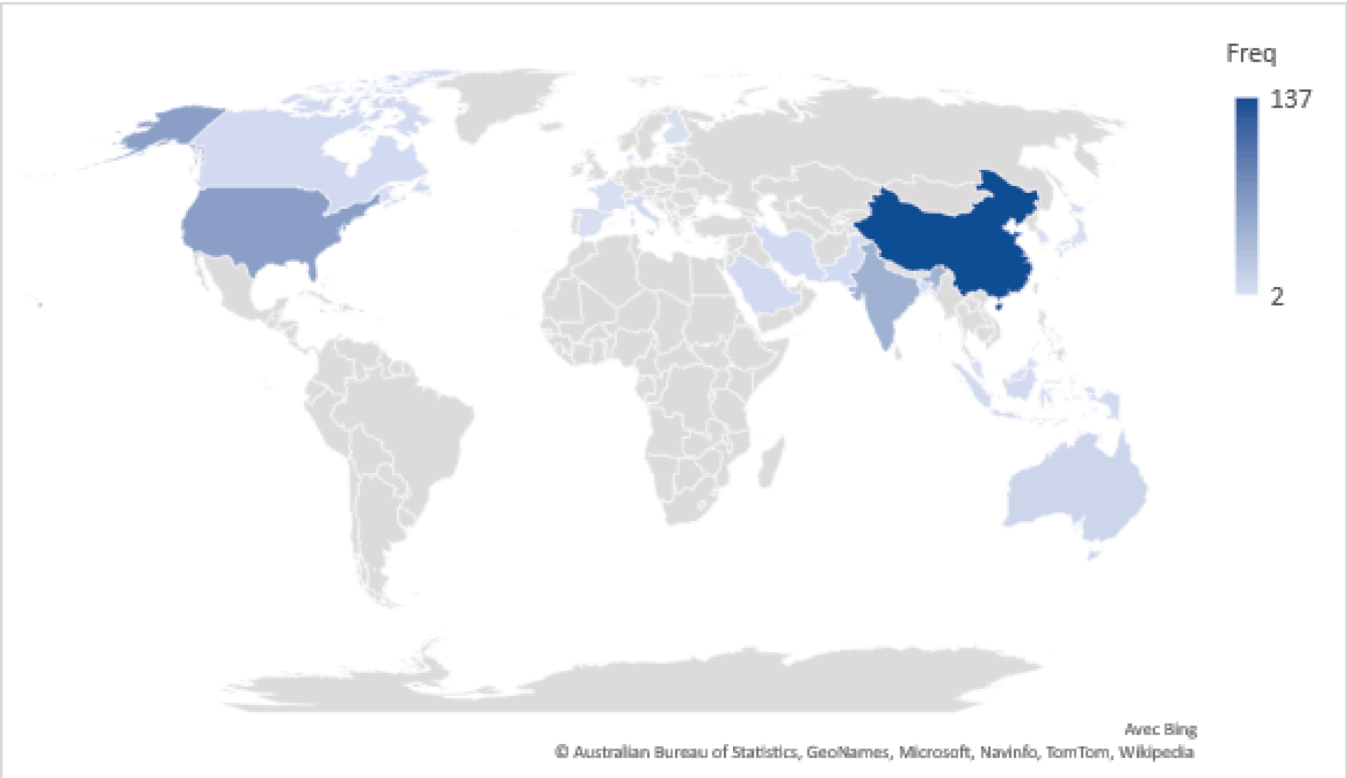


Fig. 3. Geographic distribution of countries.

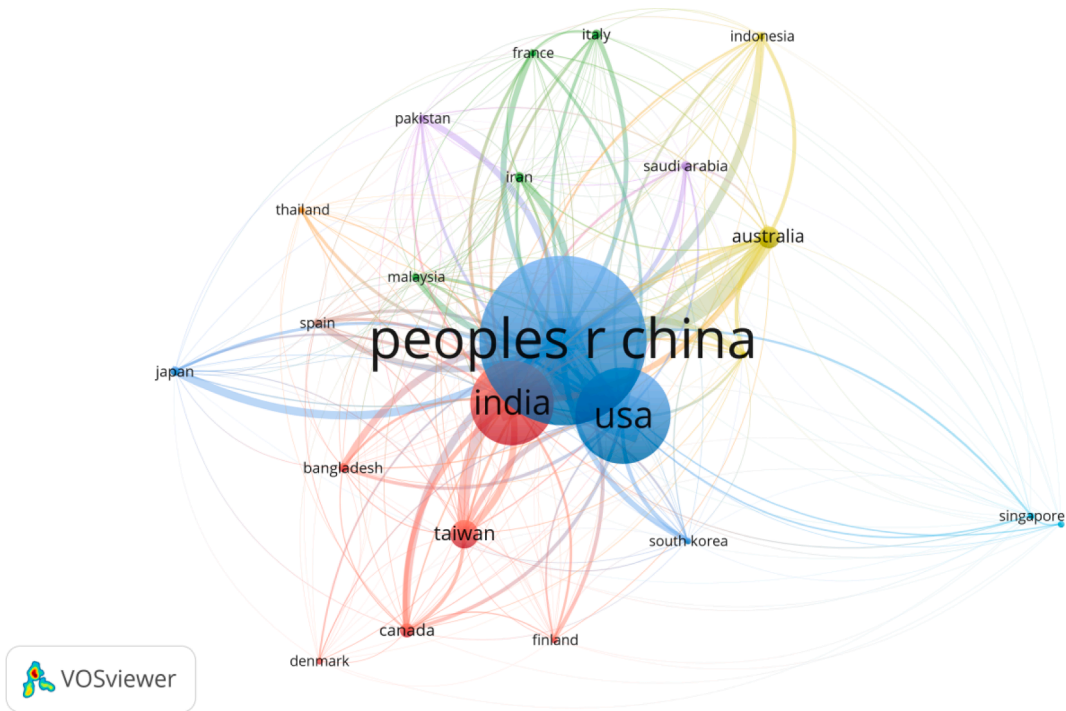


Fig. 4. Cooperation network of productive countries.

within disciplines to a certain extent. Co-citation analysis expresses the relationship among documents based on the frequency of being cited by other documents simultaneously. The analysis of clustering and key nodes in the clustering of co-cited networks can reveal the knowledge structure, knowledge base, and evolutionary characteristics of the research field, as well as the literature, which plays a key role in the

evolutionary process. In total, 368 nodes and 901 connections were obtained, as shown in Fig. 10. The analysis identified 13 knowledge clusters based on keyword co-occurrence information. The value of the parameter “silhouette” varied between –0 and 1. The silhouette value is a measure of how similar an item is to its own cluster (cohesion) compared to other clusters (Chen et al., 2006). Greater silhouette values

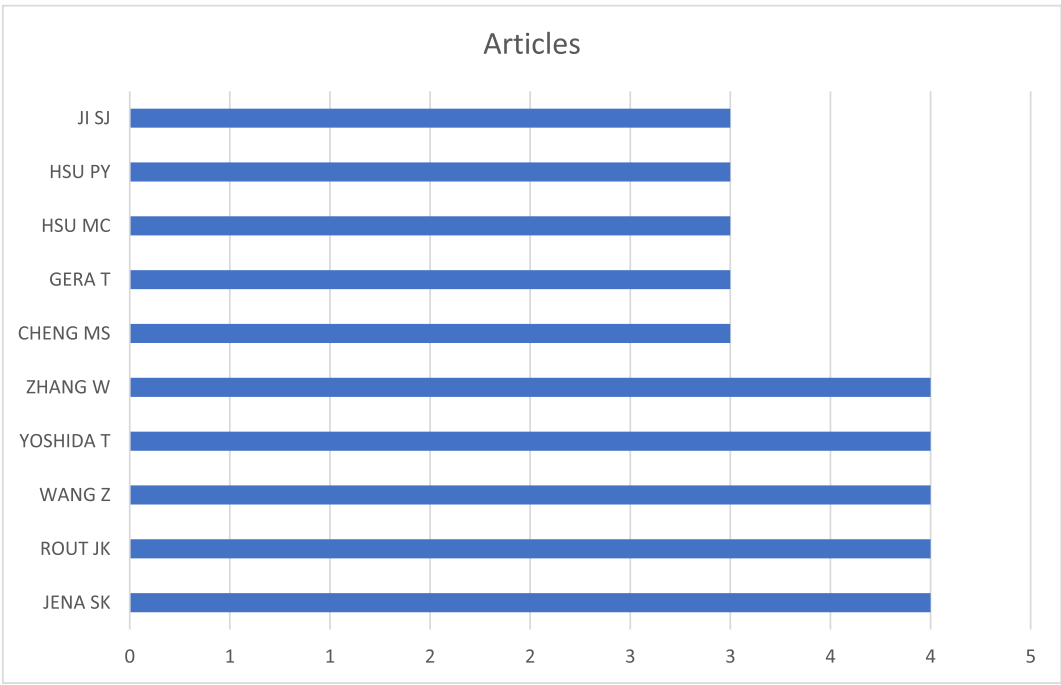


Fig. 5. Top 10 most productive authors.

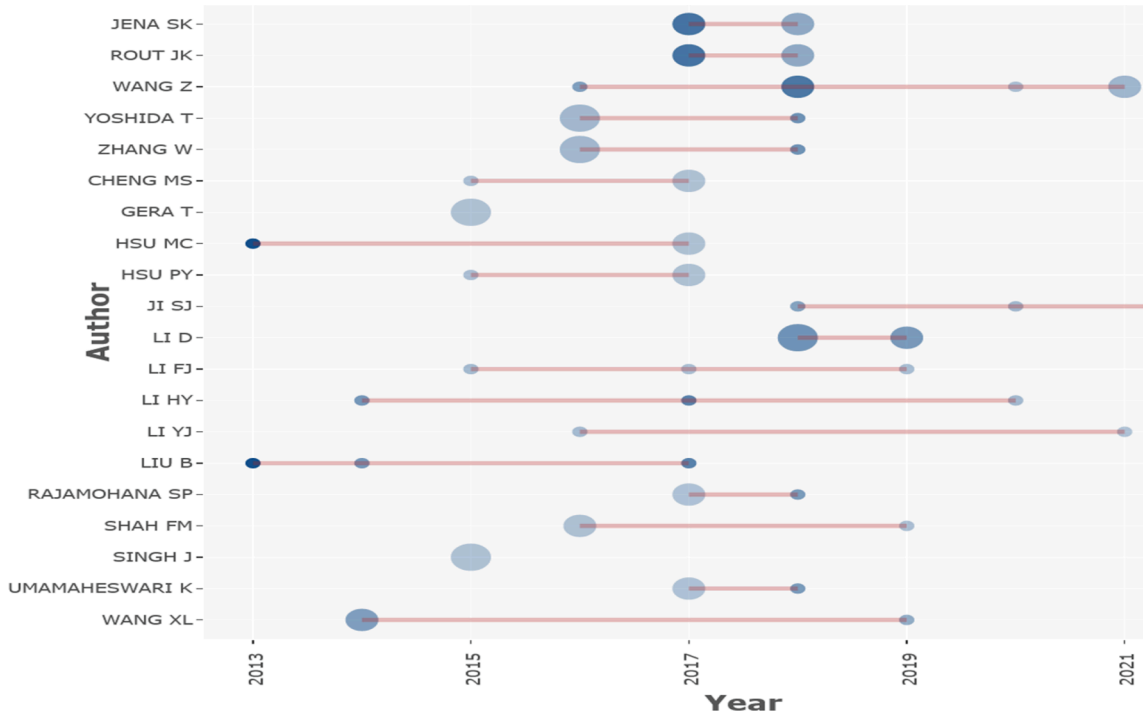


Fig. 6. Top-Authors: production over time.

suggested good-quality clustering, while lower values showed that the cluster’s components were different and scattered. The detailed information on the knowledge clusters, along with their log likelihood ratio (LLR) algorithm-based title keywords, clustering ID, size, and silhouette scores, are presented in Table 3.

In the following section, a brief analysis of the specific research groups is conducted, and the central or burst literature in each cluster is presented in the order of its respective size. We selected the top five among the 13 clusters to provide a detailed explanation.

3.2.1. Cluster 0: “Revisiting Semi-Supervised Learning” and cluster 3: “Supervised Framework”

ML models must learn from the data at hand. On the one hand, supervised learning models are trained with labeled data sets; however, labeled data can be difficult to find. Labeled data are used because when the algorithm predicts the label, the difference between the prediction and the label can be calculated and then minimized for accuracy. On the other hand, unsupervised learning does not require labeled data, as unsupervised models learn to identify patterns and trends or categorize

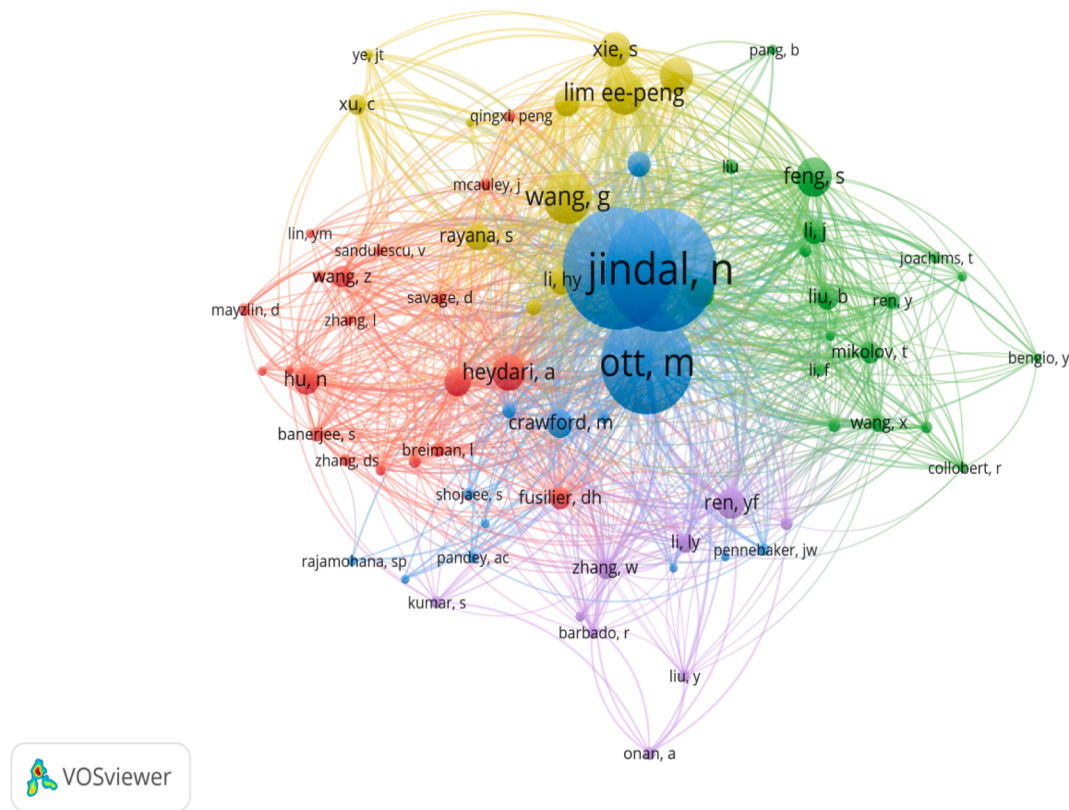


Fig. 7. Network of co-authors.

Table 2
Top 8 cited authors with the strongest citation bursts.

Cited Authors	Year	Strength	Begin	End	2012–2021
FENG S	2012	4.32	2015	2016	
LI H	2012	3.97	2016	2018	
JINDAL NITIN	2012	3.55	2015	2017	
ZHOU L	2012	3.3	2016	2017	
LI LY	2012	2.89	2019	2021	
WANG G	2012	2.83	2014	2015	
BLUM A	2012	2.45	2015	2017	
LI	2012	2.41	2018	2019	

Note: Timeline visualization derived from the expanded dataset for the top cited authors. The red bars represent frequently cited authors. The green bars represent infrequently cited authors.

data without labeling. Consequently, more data are available for unsupervised learning because most of the data are not labeled. Meanwhile, semi-supervised ML is a combination of supervised and unsupervised learning. It uses a small amount of labeled data and a large amount of unlabeled data, which provides the benefits of unsupervised and supervised learning, while avoiding the challenges of having to find a large amount of labeled data. This means that semi-supervised ML can train a model to label data without having to use so much labeled training data. Many studies have proposed semi-supervised models to detect fake reviews online.

Semi-supervised methods have been introduced in the literature to address the lack of reliable labeling of reviews. For example, Hassan and Islam (2019) used semi-supervised learning, while we used the expectation–maximization algorithm. As features, they used word frequency count, sentiment polarity, and length of review. Roobini et al. (2020) introduced some semi-supervised and supervised text mining models to detect fake online reviews and compared the efficiency of both techniques on datasets containing hotel reviews. Rout et al. (2017) explained

how semi-supervised learning methods could be used to detect spam reviews prior to demonstrating their utility using a dataset of hotel reviews.

Cluster zero, labeled “Revisiting Semi-Supervised Learning,” was the largest, containing 63 publications. The publication with the highest coverage in this cluster was that of Wu et al. (2020), who proposed an antecedent–consequence–intervention conceptual framework to develop an initial research agenda for investigating fake reviews. They identified 20 future research questions and suggested 18 propositions. We cited five of their future research questions and propositions: “Why do individual consumers post fake reviews without any external financial incentives?” “What are the underlying psychological mechanisms?” “Why do review platforms post fake reviews?” “How can we quantify the motivations for posting fake reviews?” and “Do promoters have different motivations for different kinds of products, markets, or situations; and how can this happen?”.

Cluster 3, labeled “Supervised Framework,” was the fourth largest, comprising 37 publications. The publication with the highest coverage in this cluster was that of Basiri et al. (2019), who presented a supervised framework for reviewing spam detection in the Persian language.

3.2.2. Cluster 1: “Textual-based features” and cluster 4: “opinion mining”

Text classification is a task in natural language processing (NLP) enabling the classification of free-text documents based on predefined categories. Classes can be based on topic, genre, or sentiment. The progress of NLP research has achieved enormous results, with many studies demonstrating DL methods as cutting-edge technology for executing such tasks. Opinion mining is built on sentiment analysis approaches and takes one of two forms: polarity-based, where parts of texts are classified as either positive or negative, or valence-based, where the intensity of the sentiment is considered.

Cluster 1, labeled “Textual-based Features,” contained 52 publications. The publication with the highest coverage in this cluster was that of Mohawesh et al. (2021), who detailed the task of fake review

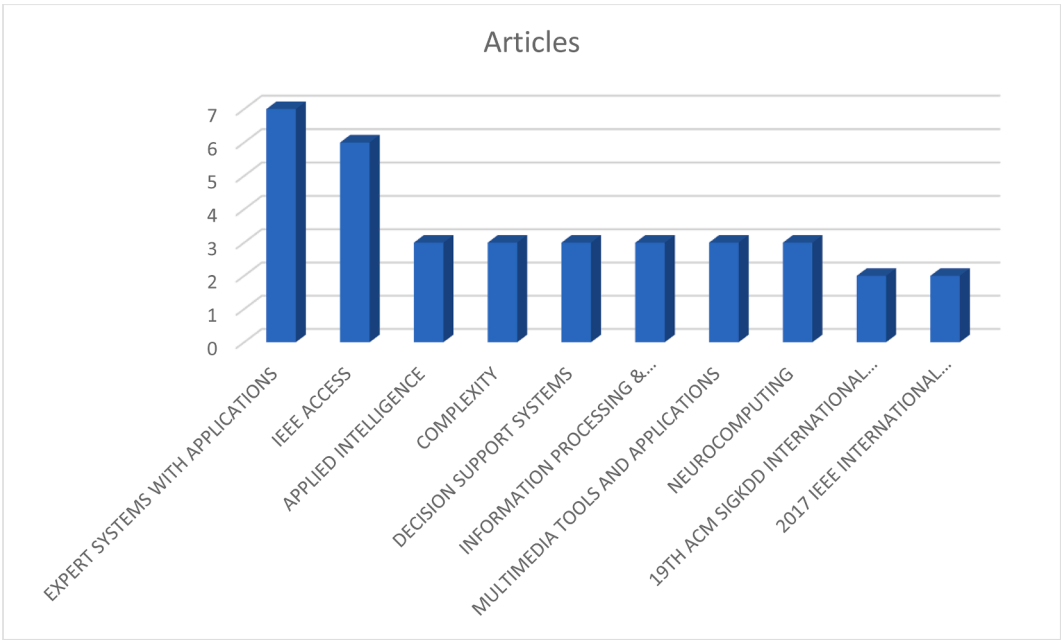


Fig. 8. Top 10 most productive journals.

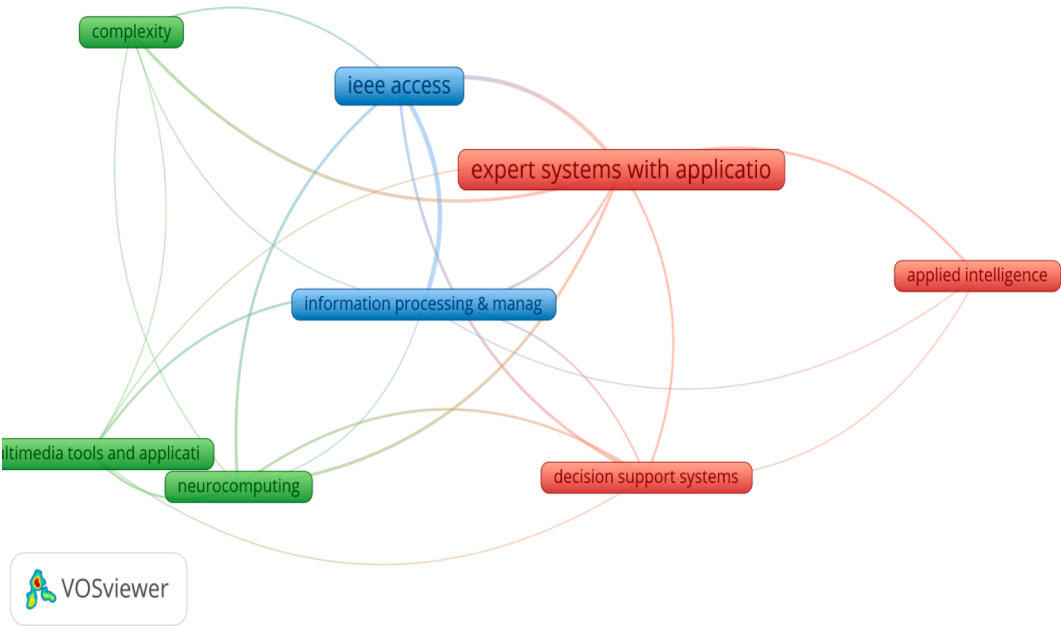


Fig. 9. Network of journals.

detection. They also critically analyzed existing feature extraction techniques and other existing methods based on two groups: traditional statistical ML and DL methods. Furthermore, they conducted a benchmark study to investigate the performance of different neural network models and transformers that have not yet been used for fake review detection.

Cluster 4, labeled “Opinion Mining,” contained 33 publications. The publication with the highest coverage in this cluster was that of [Sheibani \(2012\)](#), who presented a literature review of what has been covered in the field of spam recognition.

3.2.3. Cluster 6: “graph-based review spammer group”

Cluster 6 was labeled the “Graph-based Review Spammer Group” and comprised 29 publications. The publication with the highest

coverage in this cluster was that of [Wang et al. \(2018\)](#). They introduced a computing framework, GGSPAM, to detect review spammer groups by exploiting the topological structure of the underlying reviewer graph, which revealed co-review collusion.

4. Background

4.1. An overview

[Jindal and Liu \(2008\)](#) introduced the problem of fake review detection based on prior research involving the classification and summarization of opinions using NLP (Natural Language Processing) and data mining (web spam and email spam). They identified three types of spam reviews. The first type was termed “untruthful opinions,” also

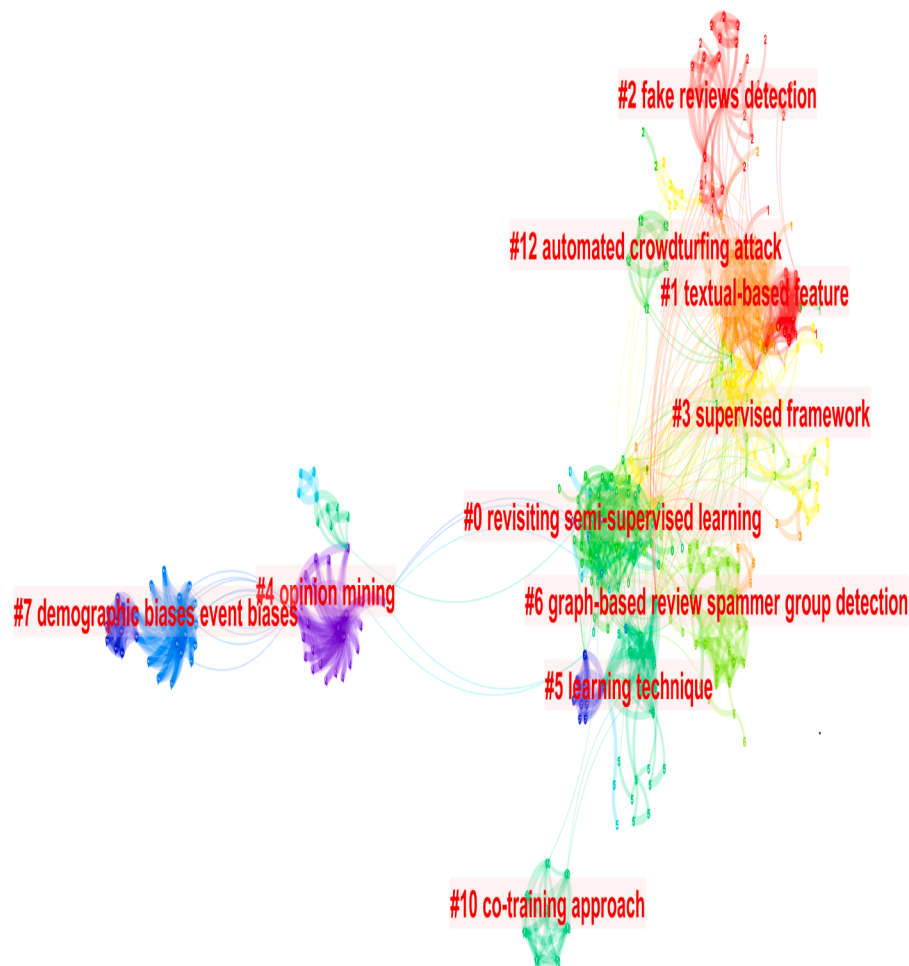


Fig. 10. Cluster map of Artificial Intelligence applications in fake review detection in the 2012–2021 period.

known as “fake reviews” or “bogus reviews.” These included reviews that purposely misled readers, opinion mining systems that gave underserving positive reviews to certain products or services to promote them (also called “hyper spam”), or reviews that gave malicious negative reviews to some other products or services to damage their reputation (also called “defaming spam”). The second type was termed “reviews on brands only,” which included those that did not comment on the products or services specifically, but only on the brands, the manufacturers, or the sellers. The third type, called “non-reviews,” included advertisements and other irrelevant reviews. Jindal and Liu (2008) manually labeled training examples for Type 2 and Type 3 spam reviews, as these are recognizable based on their content. However, they highlighted the difficulty of recognizing whether a review was an untruthful opinion spam (Type 1, fake review). Therefore, they proposed ensemble models based on unigram and bigram features. Their models demonstrated the best performance in terms of accuracy compared with naive Bayes, random forest, and support vector machines (SVMs).

4.2. The effect of fake reviews

It is well known that user-generated content includes incisive information that can be used for various applications. It affects many consumers’ buying choices, improves sellers’ ability to maintain high reputation and revenue, informs sellers about their competitors’ activities, and helps them to quickly improve and prevent any quality-of-service issues (Jindal & Liu, 2008). Thus, the growing attractiveness of online review systems encourages malicious and fraudulent intent by

enabling others to manipulate reviews to affect consumers’ buying decisions (Choi et al., 2017; Heydari et al., 2015), product and service reputations (Filieri et al., 2015; Petrescu et al., 2018), sales volumes (Dellarocas, 2006), and competition (Lee et al., 2018).

Retailers and merchants are obvious participants in the fight against fake reviews, as different types of actors seek to exploit publicly available product/service review sites to promote services, goods and brands (Heydari et al., 2015). Fake reviews trick consumers by using fake information about product/service quality, which, in turn, causes customer dissatisfaction (Lim et al., 2010). Furthermore, vendors use online review procedures as a instruments with which to damage and vilify competitors (Luca & Zervas, 2016).

The spread of fake reviews affects customer decision-making and threatens the welfare of honest traders (Ott, Cardie, & Hancock, 2013). In addition, it impacts public faith in the whole online platform (Vidyanagama et al., 2020). Thus, in their fight to preserve the interests of honest users and sellers, fake review detection has become a crucial task for review platforms. Human assessors constantly fail to differentiate fake reviews from authentic ones (Crawford et al., 2015). To determine whether a specific review is fake, an assessor cannot merely examine this review separately from other reviews of the same or other products, which makes review analysis a nontrivial and time-consuming task (Mukherjee et al., 2013). The constant progress in spamming techniques, the participation of freelancers in generating fake reviews (Yao et al., 2017), and the ongoing cooperation of spammers in groups (Rout et al., 2017) all pose serious challenges to the task of fake review detection.

Table 3
Details of knowledge clusters.

Cluster ID	Size	Silhouette	Mean (Year)	Label (LLR)
0	63	0.79	2013	revisiting semi-supervised learning; online deceptive review detection; fake opinion; supervised learning; empirical analysis
1	52	0.901	2017	textual-based feature; using machine learning classifier; fake consumer review detection; integrating word embedding
2	38	0.883	2017	fake review detection; detecting fake reviews; fine-grained aspects plan; behaviour-based featuring approach; parallel environment
3	37	0.88	2015	supervised framework; Persian language; authentic review; encoder-decoder model; probabilistic review graph.
4	33	0.97	2010	opinion mining; literature review; reputation competition; Yelp review fraud; product review
5	31	0.941	2012	learning technique; using hybrid machines; ensemble approach; using active learning; review spam
6	29	0.899	2014	graph-based review spammer group detection; lda-based group; product review; fraud de-anonymization; spamming detection
7	27	0.98	2010	demographic biases, event biases; manipulation; external validity; current method; review spam detection
10	8	0.998	2014	co-training approach; support vector machine; spam review identification; review spam detection; fake review
12	5	0.985	2012	automated crowdtrurfing attack; defences; online review system; review spam detection; fake review

Note: In terms of uniqueness and coverage, the LLR Log-likelihood ratio produces the optimal results.

5. Antecedents

Between 2010 and 2021, only 13 articles from the 181 articles published on the use of AI in fake review detection can be categorized in the discipline of business (Banerjee & Chua, 2021; Feng, Su, Li, Zheng, & Li, 2018; González-Ortega, Soyer, Insua, & Ruggeri, 2021; Kauffmann et al., 2020; Keshavarz, Waheed, Rachdi, & Alhadj, 2018; Kumar, Venugopal, Qiu, & Kumar, 2018a; Li, Meng, & Pan, 2020; Luca & Zervas, 2016; Moon, Kim, & Iacobucci, 2021; Salminen, Kandpal, Kamel, Jung, & Jansen, 2022; Satia Budhi, Chiong, Wang, & Dhakal, 2021; Saumya & Singh, 2022; Usai, Pironti, Mital, & Mejri, 2018; Wang, Li, & Yi, 2019; Yu, Zuo, Feng, An, & Chen, 2019).

Thus, the most established literature on the topic of AI applications in fake review detection is found in the field of computer science and focuses on developing data-mining algorithms that detect fake reviews. The literature in this area can be differentiated based on three criteria: the dataset used, the adopted models (models exploiting content similarity, writing footprints, behavioral footprints, network footprints, and rating footprints), and the technical detection methods (supervised and semi-supervised, convolutional neural network (CNN), and word-order-preserving CNN).

In terms of dataset, approximately 24 % of the studies used the Amazon dataset, 23 % used the Yelp dataset, and 9 % used the TripAdvisor dataset. The rest used Dianping, AMT, eBay, Apple App Store, and others.

In terms of models, early fake review detection efforts assumed it would be too complicated to distinguish fake reviews from authentic ones by analyzing how they express opinions. Thus, they focused on

detecting non-reviews and duplicate reviews (Table 4). The second model wave investigates writing footprints based on the assumption that spammers are incapable of writing reviews with similar levels of expressed emotions as real reviewers, which are reflected in the text they produce (Table 4). The third model wave exploits behavioral footprints by looking at a set of reviews authored by a user to identify anomalies in their behaviors. The fourth model wave exploits network footprints based on the assumption that dependency relationships may occur among the patterns observed in review content, reviewer behaviors, and product characteristics (Table 4). The final model wave exploits rating footprints and focuses on analyzing rating distributions for detecting spamming (Table 4).

In terms of classification algorithms, Lin et al. (2014) applied supervised and unsupervised methods based on the Sparse Additive Generative Model. They found that this achieved the best accuracy compared with SVMs in one domain. Hernández-Castañeda et al. (2017) experimented with different feature sets for training support vector networks (SVNs): a continuous semantic space model source represented by LDA topics, a word-space model, and dictionary-based features. Their

Table 4
Fake review detection: a systematic literature review.

Streams of Models	Studies	Hypothesis	Dataset
Models exploiting content similarity	Algur et al. (2010); Lai et al. (2010); Feng et al. (2012b); Li et al. (2015); Al Najada & Zhu (2014); Banerjee & Chua, (2017); Cardoso et al. (2018); Zhang et al. (2018)	It is hard to differentiate fake reviews from authentic ones	Epinion Amazon Yelp TripAdvisor Ott et al. (2011)
Models exploiting writing footprints	Li et al. (2011); Ott et al. (2011); Feng Banerjee et al. (2012a); Harris (2012); Xu & Zhao (2012); Fusilier et al. (2013); Sun et al. (2013); Li et al. (2014); Hernández Fusilier et al. (2015); Hai et al. (2016); Yao et al. (2017)	Spammers are unable to write reviews with the same level of expressed emotion as real reviewers	Amazon Dianping Ott et al. (2011) Yelp TripAdvisor
Models exploiting behavioral footprints	Lin et al. (2014); Zhang et al. (2016); Wang et al. (2012); Lu et al. (2013); Mukherjee et al. (2012); Savage et al. (2015); Kumar et al. (2018b); Barbado et al. (2019)	Analyzing a whole set of reviews created by a user can show abnormalities in their behavior	Resellerratings Amazon AMT Yelp
Models exploiting network footprints	Akoglu et al. (2013); Fei et al. (2013); Xu et al. (2013); Fayazi et al. (2015); Rayana & Akoglu (2015); Xue et al. (2015); Shah et al. (2016); (Kumar et al., 2018a)	Dependency relationships exist between the review content, reviewer behavior and product characteristics	Amazon Yelp SWM eBay
Models exploiting rating footprints	Xie et al. (2012); Li et al. (2014); Heydari et al. (2016); Hooi et al. (2016); Mukherjee et al. (2016)	Typical distributional patterns in the ratings submitted by honest reviews, permitting abnormality detection	TripAdvisor Resellerratings Dianping Amazon FlipKart SWM Yelp Apple App Store

results demonstrated the efficiency of SVN in detecting fake reviews in one domain (average accuracy: 86 %), mixed domains (75 %), and cross-domains (52 %–64 %) on combined features.

Another group of researchers on fake review detection proposed DL models. DL techniques can extract latent data representation, which contributes to enhancing fake review detection performance. For example, Li et al. (2015) built a fake review detector based on DL by applying CNN. They used the document representation concept, in which each review was translated into a word vector for training and testing the model. Their results demonstrated that the proposed model was effective in detecting cross-domain fake reviews. Moreover, their suggested model outperformed long short-term memory (LSTM) in mixed-domain fake reviews. Similarly, Ren and Ji (2017) explored a neural network model for learning document-level representation to detect deceptive opinion spam. Deceptive opinion spam is a type of review with fabricated opinions intentionally written to look authentic. They found that the neural model performance surpassed previous works that used a discrete baseline. Indeed, the neural model can properly capture nonlocal discourse information over sentence vectors. Furthermore, with an attention mechanism, the neural model outperformed the model without an attention mechanism.

Meanwhile, Zhao et al. (2018) also optimized the CNN model by introducing word-order characteristics in the model's convolution and pooling layers. Given that prior works were based on textual semantics and emotional polarity, they proposed a word-order-preserving CNN (OPCNN) model to maintain word-order characteristics during the process of opinion analysis. After conducting experiments on the "TensorFlow" open-source deep learning platform, their results proved that the proposed OPCNN detection model was better than the CNN-based baseline.

However, the effectiveness of CNN and OPCNN depended on a careful selection of the window size (Lai et al., 2015). Moreover, it has been shown that CNN lacked the ability to capture the overall semantic information in long texts (Liu et al., 2020). Lai et al. (2015) proposed a recurrent neural network (RNN) as an alternative to capture contextual information when learning word representations, which might introduce less noise compared to traditional window-based neural networks. In addition, to capture the key components in the texts, they used a max-pooling layer that automatically judged which words played key roles in text classification. Their experimental results revealed that the proposed method outperformed the ML algorithms with unigram and bigram baselines for text classification proposed by Wang and Manning (2012); the baseline of Huang et al. (2012), which used the weighted average of the word embeddings and subsequently applied a SoftMax layer with CNN (Collobert et al., 2011); with Tree Kernel (Post & Bergsma, 2013); with LDA (Hingmire et al., 2013); with RNNs (Socher et al., 2011); and with recursive neural tensor networks (Socher et al., 2013).

Based on the abovementioned information, RNN may suffer greatly from vanishing and exploding gradients, which can hamper the learning of long data sequences. These two problems belong to the class of open problems in ML, and research to resolve them is ongoing. For example, LSTM architecture has been designed to address vanishing and exploding gradients in conventional RNNs (Sak et al., 2014).

Liu et al. (2020) used a neural network based on the bidirectional LSTM algorithm (BiLSTM) to learn document-level representation of reviews. They then incorporated the feature representation into BiLSTM (BiLSTMWF) and conducted cross-domain and mixed-domain experiments to prove the effectiveness of their proposed method. The domain migration experiment on the cross-domain dataset demonstrated that the proposed model was more robust than other neural network-based methods. Indeed, the domain-independent experiment on the mixed-domain dataset revealed that their model outperformed the most advanced method. In particular, their results surpassed those of other strong baseline methods, with the highest F1 value of 87.6 %.

6. Avenues for future research

Following classic review articles (Basu et al., 2022; Billore & Anisimova, 2021; Khattoon & Rehman, 2021; Nanda & Banerjee, 2021; Södergren, 2021), we provide detailed directions for future research. For over a decade, there have been enormous advances in our knowledge of fake review detection, with most research in this domain focusing on fake review content analysis. The timelines of co-citation clusters from 2010 to 2021 (Fig. 11) show that the textual-based features using ML classifiers indeed exhibit this global trend. In addition, current research is mostly centered on the semi-supervised ML model using a recurrent convolutional neural network (RCNN). Based on the timelines of the co-citation clusters and the 4Ws, we identified some challenges that must be addressed in future research. First, spammers develop their methods to simulate the behavior of honest reviewers, making fake review detection techniques less effective. Second, the majority of fake review detection methods are responsive and cannot operate in real time. Third, the majority of methods are only compatible with a particular platform. Fourth, all these methods fail to differentiate between fake and honest reviews, which leads to the unnecessary provocation of honest reviewers. Thus, based on these challenges, we developed five propositions for future research.

6.1. Spammer detection group

Thus far, numerous studies have focused on individual review spam detection, but recently, a trend toward detecting group spammers has emerged. These groups of "reviewers" (groups of spammers) who work collaboratively to write fake reviews are more damaging, as they can take total control of the sentiment on the target product due to their size. According to Mukherjee et al. (2012), detecting a group of spammers is an essential part of fake review detection. Such a group often starts the propagation of fake reviews in limited real-time intervals. Hence, by considering studies of burst patterns to detect fake reviews, Li et al. (2017) found high accuracy in fake review detection. Therefore, the study of burst patterns using new techniques to detect group spammers must be further investigated in future research. Future studies must also focus on understanding the strategies and behaviors of group spammers. The suggested models must identify both fake product reviews and spammers by creating specific spam indicators.

We furthermore suggest that future researchers should address the question: "How can group spammers manipulate customers' decisions?" We suggest using the transferable belief model (TBM) to answer this question. The belief function theory was initially proposed by Dempster (2008), subsequently validated by Shafer (1976), and axiomatically explained by Smets and Kennes (2008) in a TBM. The TBM consists of two levels: a credal level, wherein beliefs are entertained, and a pignistic degree, wherein beliefs are employed in order to make decisions.

6.2. Explainable ML fake review detection model

ML has played a substantial role in NLP, with tremendous results. It is nevertheless considered a "black box" that does not have declarative knowledge for further, more in-depth explanations of results. None of the DL models for fake review detection are interpretable. Consequently, it is difficult to trust the model's performance and results. Thus, in response to the "black box" problem, several techniques have recently been proposed for interpreting the predictions of complex models. Lundberg and Lee (2017), for example, proposed a unified framework for interpreting predictions, called SHapley Additive exPlanations (SHAP). Ribeiro et al. (2016) proposed a local interpretable model-agnostic explanations (LIME) technique, which explains the predictions of any classifier in an interpretable way by learning an interpretable model locally around the prediction. Friedman (2001) proposed a partial dependence plot (PDP), which could demonstrate whether the relationship between the target and a feature is linear,

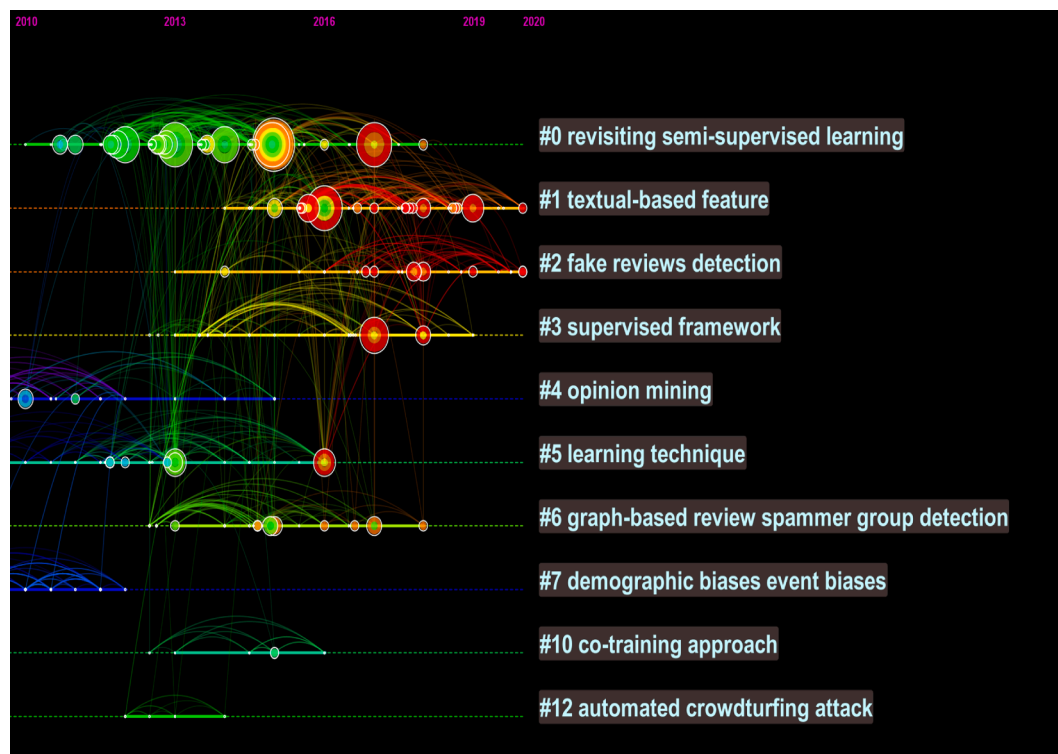


Fig. 11. Timelines of co-citation clusters from 2010 to 2021. The color of the line between the nodes represents the time when the relevant publication appeared, where a dark color represents earlier years and a light color represents later years. Different colored lines indicate that the two articles were referenced in the same article.

monotonic, or more complex. Based on three criteria, Jabeur et al. (2021) found that SHAP was a superior tool compared to the others. The three decisive criteria, namely, overall interpretability, local interpretability, and SHAP values, are calculable for any tree model, whereas other methods used linear regression or logistic regression models as alternatives. Therefore, future research must respond to the need for explainable fake review detection models.

Explainable ML builds user trust, provides information on improving a model, and improves understanding of the modeled process. We propose herein the use of Truth-Default Theory (TDT) with explainable ML methods to facilitate a sufficient understanding of the factors that push individuals to create fake reviews. Levine (2014) also proposed the TDT as a new theory of deception and deception detection, defining the former as the act of “intentionally, knowingly, and/or purposely misleading another person (p. 2)” TDT provides an overarching logical structure that ties together various models, including the Prolific Liars Model, the Deception Motives Module, the Projected Motive Model, the Veracity Effect, and the Park–Levine Probability Model, into a coherent theory.

6.3. Handling the vanishing and exploding gradient problems

As presented in the literature review, RNNs are better than CNNs due to the former’s ability to capture contextual information when learning word representations, which introduces less noise compared to traditional window-based neural networks. The experimental results showed that RNNs outperformed other ML algorithms that used unigrams and bigrams for text classification, weighted average of the word, CNN, Tree Kernel, LDA, Recursive NN, and Recursive Neural Tensor Networks. Therefore, RNNs may suffer greatly from the vanishing and exploding gradient problems, which hamper the learning of long data sequences. Several researchers have proposed various techniques to resolve the RNN vanishing and exploding gradients problem. For example, Ribeiro et al. (2020) reformulated the problem of exploding gradients in terms of

cost function smoothness and explained the distinction between vanishing gradients and the need for the RNN to teach attractors to fully use its expressive power. Based on this assumption, they proposed stable RNNs and unitary (or orthogonal) RNNs as solutions. Kag et al. (2019) proposed an incremental RNN (iRNN), a novel RNN architecture that resolves the exploding/vanishing gradient problem. They demonstrated how their method did not suffer from performance degradation. After the experiments, the iRNN showed competitive performance against standard LSTM networks on long-term-dependency (ltd) and other non-ltd tasks. Thus, future studies must take up the opportunity for continuous work to improve RNN and build new and more efficient fake review detectors.

6.4. Multilingual deceptive and fake review detection for group spammers

Fake reviews can be written in different languages (English, French, Chinese, Persian, or Arabic, for example). However, thus far, only a few studies have utilized multilanguage fake review datasets (Capuozzo et al., 2020; Mukherjee et al., 2013). As group spammers initiate the propagation of fake reviews in limited real-time intervals, they copy text from other datasets. They can also use applications to translate English reviews into any other language. Therefore, future research must focus on the detection of multilingual fake reviews.

7. Conclusion

This paper presents an extensive bibliometric analysis of AI applications in fake review detection. First, we presented important research to date on ML-based fake review detection. We then applied a bibliometric approach to better identify the main advances and new perspectives in this field and to highlight the development, hot spots, and trend directions for future research on this subject.

The results demonstrate an increasing research interest in AI applications for the detection of fake reviews. In particular, the number of

publications increased gradually between 2012 and 2021, with 22 countries and 518 authors contributing to 181 publications on AI applications deployed in fake review detection. Based on the analysis of highly cited references and the quantitative reasoning of keyword frequency and centrality, we conclude that the research hotspots are centered on the semi-supervised ML model using RCNN, RNN, and other algorithms for fake review detection. Textual-based features using ML classifiers also present a future research trend on applying AI for detecting fake reviews. Some possible directions for future research are determined by combining keyword burst analysis with a critical reading of influential references.

The major contribution of this study is the classification of research articles to integrate the current state of literature on this topic and to understand the role of AI in fake review detection. AI and ML are two closely linked technologies that are becoming more popular in academia in the field of marketing. However, despite the crucial need to support marketing academia to increase our understanding of the consequences of AI and ML on marketing systems, no study has provided a comprehensive analysis of this research to date. Thus, our study addresses this gap in the research and helps decision-makers understand the threat of group spammers, motivating them to improve fake review regulation frameworks. Review platforms should also strengthen pre-review procedures to prevent the proliferation of fake reviews.

CRediT authorship contribution statement

Sami Ben Jabeur: Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hossein Bal-louk:** Writing – original draft, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Wissal Ben Arfi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Conceptualization. **Jean Michel Sahut:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Ahmed, H., Traore, I., & Saad, S. (2018). Detecting opinion spams and fake news using text classification. *Security and Privacy*, 1(1), e9.
- Akoglu, L., Chandy, R., & Faloutsos, C. (2013). Opinion Fraud Detection in Online Reviews by Network Effects. *Proceedings of the International AAAI Conference on Web and Social Media*, 7(1), Article 1.
- Algur, S. P., Patil, A. P., Hiremath, P. S., & Shivashankar, S. (2010). Conceptual level similarity measure based review spam detection. *International Conference on Signal and Image Processing*, 2010, 416–423. <https://doi.org/10.1109/ICSIP.2010.5697509>
- Aria, M., & Cuccurullo, C. (2017). bibliometrix: An R-tool for comprehensive science mapping analysis. *Journal of Informetrics*, 11(4), 959–975. <https://doi.org/10.1016/j.joi.2017.08.007>
- Bahoo, S., Alon, I., & Paltrinieri, A. (2020). Corruption in international business: A review and research agenda. *International Business Review*, 29(4), Article 101660.
- Ballouk, H., Jabeur, S. B., & Arfi, W. B. (2021). Analyse bibliométrique de la littérature portant sur la co-thématique: Responsabilité sociale des entreprises et développement durable. *Question (s) de Management*, 36(6), 87–99.
- Banerjee, S., & Chua, A. Y. K. (2017). Authentic versus fictitious online reviews: A textual analysis across luxury, budget, and mid-range hotels. *Journal of Information Science*, 43(1), 122–134. <https://doi.org/10.1177/0165551515625027>
- Banerjee, S., & Chua, A. Y. K. (2021). Calling out fake online reviews through robust epistemic belief. *Information & Management*, 58(3), Article 103445. <https://doi.org/10.1016/j.im.2021.103445>
- Barbado, R., Araque, O., & Iglesias, C. A. (2019). A framework for fake review detection in online consumer electronics retailers. *Information Processing & Management*, 56(4), 1234–1244. <https://doi.org/10.1016/j.ipm.2019.03.002>
- Basiri, M. E., Safarian, N., & Farsani, H. K. (2019). A Supervised Framework for Review Spam Detection in the Persian Language. *2019 5th International Conference on Web Research (ICWR)*, 203–207. <https://doi.org/10.1109/ICWR.2019.8765275>
- Basu, R., Paul, J., & Singh, K. (2022). Visual merchandising and store atmospherics: An integrated review and future research directions. *Journal of Business Research*, 151, 397–408. <https://doi.org/10.1016/j.jbusres.2022.07.019>
- Billore, S., & Anisimova, T. (2021). Panic buying research: A systematic literature review and future research agenda. *International Journal of Consumer Studies*, 45(4), 777–804. <https://doi.org/10.1111/ijcs.12669>
- Butler, L., & Visser, M. S. (2006). Extending citation analysis to non-source items. *Scientometrics*, 66(2), 327–343. <https://doi.org/10.1007/s11192-006-0024-1>
- Butler, T., Feller, J., Pope, A., Emerson, B., & Murphy, C. (2008). Designing a core IT artefact for Knowledge Management Systems using participatory action research in a government and a non-government organisation. *The Journal of Strategic Information Systems*, 17(4), 249–267. <https://doi.org/10.1016/j.jsis.2007.10.002>
- Cao, Q., Duan, W., & Gan, Q. (2011). Exploring determinants of voting for the “helpfulness” of online user reviews: A text mining approach. *Decision Support Systems*, 50(2), 511–521. <https://doi.org/10.1016/j.dss.2010.11.009>
- Mukherjee, A., Venkataraman, V., Liu, B., & Glance, N. (2013). What Yelp Fake Review Filter might be Doing? A Case Study on Commercial vs. Crowdsourced Fake Reviews. *Proceedings of the 7th International AAAI Conference on Weblogs and Social Media*.
- Lai, K., Xu, Q., Lau, R.Y.K., Li, Y., & Jing, L. (2010). Toward a Language Modeling Approach for Consumer Review Spam Detection. *2010 IEEE 7th International Conference on E-Business Engineering* (pp. 1–8). <https://doi.org/10.1109/ICEBE.2010.47>
- Capuzzo, P., Lauriola, I., Strapparava, C., Aioli, F., & Sartori, G. (2020). DecOp: A Multilingual and Multi-domain Corpus For Detecting Deception In Typed Text. *Proceedings of the 12th Language Resources and Evaluation Conference*, 1423–1430. <https://aclanthology.org/2020.lrec.1-178>
- Cardoso, E. F., Silva, R. M., & Almeida, T. A. (2018). Towards automatic filtering of fake reviews. *Neurocomputing*, 309, 106–116. <https://doi.org/10.1016/j.neucom.2018.04.074>
- Cardy, M. (2015, October 18). Amazon targets 1,114 “fake reviewers” in Seattle lawsuit. *BBC News*. <https://www.bbc.com/news/technology-34565631>
- Casadesus-Masanell, R., & Ricart, J. E. (2011). How to design a winning business model. *Harvard Business Review*, 89(1/2), 100–107.
- Chakma, R., Paul, J., & Dhir, S. (2021). Organizational ambidexterity: A review and research agenda. *IEEE Transactions on Engineering Management*, 1–17. <https://doi.org/10.1109/TEM.2021.3114609>
- Chen, X., He, Z., Keller, J. M., Anderson, D., & Skubic, M. (2006). Adaptive silhouette extraction in dynamic environments using fuzzy logic. 236–243.
- Choi, S., Mattila, A. S., Hoof, H. B. V., & Quadri-Felitti, D. (2017). The role of power and incentives in inducing fake reviews in the tourism industry. *Journal of Travel Research*, 56(8), 975–987. <https://doi.org/10.1177/0047287516677168>
- Collobert, R., Weston, J., Bottou, L., Karlen, M., Kavukcuoglu, K., & Kuksa, P. (2011). Natural language processing (almost) from scratch. *Journal of Machine Learning Research*, 12(ARTICLE), 2493–2537.
- Crawford, M., Khoshgoftaar, T. M., Prusa, J. D., Richter, A. N., & Al Najada, H. (2015). Survey of review spam detection using machine learning techniques. *Journal of Big Data*, 2(1), 1–24.
- Culnan, M. J., O'Reilly, C. A., III, & Chatman, J. A. (1990). Intellectual structure of research in organizational behavior, 1972–1984: A cocitation analysis. *Journal of the American Society for Information Science*, 41(6), 453–458. [https://doi.org/10.1002/\(SICI\)1097-4571\(199009\)41:6<453::AID-AS113>3.0.CO;2-E](https://doi.org/10.1002/(SICI)1097-4571(199009)41:6<453::AID-AS113>3.0.CO;2-E)
- Dabić, M., Vlačić, B., Paul, J., Dana, L.-P., Sahasranamam, S., & Glinka, B. (2020). Immigrant entrepreneurship: A review and research agenda. *Journal of Business Research*, 113, 25–38.
- Dellarocas, C. (2006). Strategic manipulation of internet opinion forums: Implications for consumers and firms. *Management Science*, 52(10), 1577–1593.
- Dempster, A. P. (2008). Upper and Lower Probabilities Induced by a Multivalued Mapping. In R. R. Yager, & L. Liu (Eds.), *Classic Works of the Dempster-Shafer Theory of Belief Functions* (pp. 57–72). Springer. https://doi.org/10.1007/978-3-540-44792-4_3
- Fang, Y., Wang, H., Zhao, L., Yu, F., & Wang, C. (2020). Dynamic knowledge graph based fake-review detection. *Applied Intelligence*, 50(12), 4281–4295. <https://doi.org/10.1007/s10489-020-01761-w>
- Fayazi, A., Lee, K., Caverlee, J., & Squicciarini, A. (2015). Uncovering Crowdsourced Manipulation of Online Reviews. *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 233–242. <https://doi.org/10.1145/2766462.2767742>
- Fei, G., Mukherjee, A., Liu, B., Hsu, M., Castellanos, M., & Ghosh, R. (2013). Exploiting burstiness in reviews for review spammer detection. *Seventh International AAAI Conference on Weblogs and Social Media*.
- Feng, N., Su, Z., Li, D., Zheng, C., & Li, M. (2018). Effects of review spam in a firm-initiated virtual brand community: Evidence from smartphone customers. *Information & Management*, 55(8), 1061–1070. <https://doi.org/10.1016/j.im.2018.05.012>
- Feng, S., Banerjee, R., & Choi, Y. (2012). Syntactic Stylometry for Deception Detection. *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 171–175. <https://aclanthology.org/P12-2034>
- Feng, S., Xing, L., Gogar, A., & Choi, Y. (2012). Distributional Footprints of Deceptive Product Reviews. *Proceedings of the International AAAI Conference on Web and Social Media*, 6(1), Article 1.
- Filieri, R., Alguezaui, S., & McLeay, F. (2015). Why do travelers trust TripAdvisor? Antecedents of trust towards consumer-generated media and its influence on recommendation adoption and word of mouth. *Tourism Management*, 51, 174–185.

- Forlano, C., De Bernardi, P., & Yahiaoui, D. (2021). Entrepreneurial universities: A bibliometric analysis within the business and management domains. *Technological Forecasting and Social Change*, 165, Article 120522. <https://doi.org/10.1016/j.techfore.2020.120522>
- Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *The Annals of Statistics*, 29(5), 1189–1232. JSTOR.
- Fusilier, D.H., Guzmán Cabrera, R., Montes-y-Gómez, M., & Rosso, P. (2013). Using PU-Learning to Detect Deceptive Opinion Spam. *Proceedings of the 4th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 38–45. <https://aclanthology.org/W13.1606>
- Gilal, F. G., Zhang, J., Paul, J., & Gilal, N. G. (2019). The role of self-determination theory in marketing science: An integrative review and agenda for research. *European Management Journal*, 37(1), 29–44.
- Glänzel, W., Leta, J., & Thijs, B. (2006). Science in Brazil. Part 1: A macro-level comparative study. *Scientometrics*, 67(1), 67–86. <https://doi.org/10.1007/s11192-006-0055-7>
- González-Ortega, J., Soyer, R., Insua, D. R., & Ruggeri, F. (2021). An adversarial risk analysis framework for batch acceptance problems. *Decision Analysis*, 18(1), 25–40. <https://doi.org/10.1287/deca.2020.0420>
- Goodell, J. W., Kumar, S., Lim, W. M., & Pattanaik, D. (2021). Artificial intelligence and machine learning in finance: Identifying foundations, themes, and research clusters from bibliometric analysis. *Journal of Behavioral and Experimental Finance*, 32, Article 100577. <https://doi.org/10.1016/j.jbef.2021.100577>
- Al Najada, H., & Zhu, X. (2014). ISRD: Spam review detection with imbalanced data distributions. *Proceedings of the 2014 IEEE 15th International Conference on Information Reuse and Integration (IEEE IRI 2014)* (pp. 553–560). <https://doi.org/10.1109/IRI.2014.7051938>
- Hai, Z., Zhao, P., Cheng, P., Yang, P., Li, X.-L., & Li, G. (2016). Deceptive Review Spam Detection via Exploiting Task Relatedness and Unlabeled Data. *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, 1817–1826. <https://doi.org/10.18653/v1/D16-1187>
- Harris, C.G. (2012). Detecting Deceptive Opinion Spam Using Human Computation. *Workshops at the Twenty-Sixth AAAI Conference on Artificial Intelligence*. <https://www.aaai.org/ocs/index.php/WS/AAAIW12/paper/view/5256>
- Hassan, R., & Islam, Md. R. (2019). Detection of fake online reviews using semi-supervised and supervised learning. *2019 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 1–5. <https://doi.org/10.1109/ECACE.2019.8679186>
- Hernández Fusilier, D., Montes-y-Gómez, M., Rosso, P., & Guzmán Cabrera, R. (2015). Detecting positive and negative deceptive opinions using PU-learning. *Information Processing & Management*, 51(4), 433–443. <https://doi.org/10.1016/j.ipm.2014.11.001>
- Hernández-Castañeda, Á., Calvo, H., Gelbukh, A., & Flores, J. J. G. (2017). Cross-domain deception detection using support vector networks. *Soft Computing*, 21(3), 585–595. <https://doi.org/10.1007/s00500-016-2409-2>
- Heydari, A., Ali Tavakoli, M., Salim, N., & Heydari, Z. (2015). Detection of review spam: A survey. *Expert Systems with Applications*, 42(7), 3634–3642.
- Heydari, A., Tavakoli, M., & Salim, N. (2016). Detection of fake opinions using time series. *Expert Systems with Applications*, 58, 83–92. <https://doi.org/10.1016/j.eswa.2016.03.020>
- Hingmire, S., Chougule, S., Palshikar, G. K., & Chakraborti, S. (2013). Document classification by topic labeling. *Proceedings of the 36th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 877–880. <https://doi.org/10.1145/2484028.2484140>
- Hooi, B., Shah, N., Beutel, A., Günnemann, S., Akoglu, L., Kumar, M., Makhija, D., & Faloutsos, C. (2016). BIRDNEST: Bayesian Inference for Ratings-Fraud Detection. *Proceedings of the 2016 SIAM International Conference on Data Mining (SDM)* (pp. 495–503). Society for Industrial and Applied Mathematics. <https://doi.org/10.1137/1.9781611974348.56>
- Hota, P. K., Subramanian, B., & Narayanamurthy, G. (2020). Mapping the intellectual structure of social entrepreneurship research: a citation/co-citation analysis. *Journal of Business Ethics*, 166(1), 89–114. <https://doi.org/10.1007/s10551-019-04129-4>
- Huang, E. H., Socher, R., Manning, C. D., & Ng, A. Y. (2012). Improving word representations via global context and multiple word prototypes. *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 873–882.
- Jabeur, S. B., Ballouk, H., Arfi, W. B., & Khalfaoui, R. (2021). Machine learning-based modeling of the environmental degradation, institutional quality, and economic growth. *Environmental Modeling & Assessment*. <https://doi.org/10.1007/s10666-021-09807-0>
- Jindal, N., & Liu, B. (2008). Opinion Spam and Analysis. *Proceedings of the 2008 International Conference on Web Search and Data Mining*, 219–230. <https://doi.org/10.1145/1341531.1341560>
- Junquera, B., & Mitre, M. (2007). Value of bibliometric analysis for research policy: A case study of Spanish research into innovation and technology management. *Scientometrics*, 71(3), 443–454. <https://doi.org/10.1007/s11192-007-1689-9>
- Kag, A., Zhang, Z., & Saligrama, V. (2019, September 25). RNNs Incrementally Evolving on an Equilibrium Manifold: A Panacea for Vanishing and Exploding Gradients? *International Conference on Learning Representations*. <https://openreview.net/forum?id=HylpqA4FwS>
- Kahiya, E. T. (2018). Five decades of research on export barriers: Review and future directions. *International Business Review*, 1172–1188.
- Kauffman, E., Peral, J., Gil, D., Ferrández, A., Sellers, R., & Mora, H. (2020). A framework for big data analytics in commercial social networks: A case study on sentiment analysis and fake review detection for marketing decision-making. *Industrial Marketing Management*, 90, 523–537.
- Keshavarz, F., Waheed, A. A., Rachdi, B., & Alhajj, R. (2018). Review spam detection by highlighting potential spammers and diminishing their effect. *International Journal of E-Business Research (IJEER)*, 14(1), 54–76.
- Khatoun, S., & Rehman, V. (2021). Negative emotions in consumer brand relationship: A review and future research agenda. *International Journal of Consumer Studies*, 45(4), 719–749. <https://doi.org/10.1111/ijcs.12665>
- Knoll, J., & Matthes, J. (2017). The effectiveness of celebrity endorsements: A meta-analysis. *Journal of the Academy of Marketing Science*, 45(1), 55–75.
- Kumar, N., Venugopal, D., Qiu, L., & Kumar, S. (2018). Detecting review manipulation on online platforms with hierarchical supervised learning. *Journal of Management Information Systems*, 35(1), 350–380. <https://doi.org/10.1080/07421222.2018.1440758>
- Kumar, S., Hooi, B., Makhija, D., Kumar, M., Faloutsos, C., & Subrahmanian, V.S. (2018b). REV2: Fraudulent User Prediction in Rating Platforms. *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining* (pp. 333–341). <https://doi.org/10.1145/3159652.3159729>
- Lai, S., Xu, L., Liu, K., & Zhao, J. (2015). Recurrent Convolutional Neural Networks for Text Classification. *Twenty-Ninth AAAI Conference on Artificial Intelligence*.
- Lee, S., Qiu, L., & Whinston, A. (2018). Sentiment manipulation in online platforms: An analysis of movie tweets. *Production and Operations Management*, 27(3), 393–416.
- Levine, T. R. (2014). Truth-default theory (TDT): A theory of human deception and deception detection. *Journal of Language and Social Psychology*, 33(4), 378–392. <https://doi.org/10.1177/0261927X14535916>
- Li, F. H., Huang, M., Yang, Y., & Zhu, X. (2011, June 29). Learning to Identify Review Spam. *Twenty-Second International Joint Conference on Artificial Intelligence*. <https://www.aaai.org/ocs/index.php/IJCAI/IJCAI11/paper/view/3097>
- Li, H., Chen, Z., Liu, B., Wei, X., & Shao, J. (2014). Spotting fake reviews via collective positive-unlabeled learning. *IEEE International Conference on Data Mining, 2014*, 899–904. <https://doi.org/10.1109/ICDM.2014.47>
- Li, H., Fei, G., Wang, S., Liu, B., Shao, W., Mukherjee, A., & Shao, J. (2017). Bimodal Distribution and Co-Bursting in Review Spam Detection. *Proceedings of the 26th International Conference on World Wide Web*, 1063–1072. <https://doi.org/10.1145/3038912.3052582>
- Li, H., Liu, B., Mukherjee, A., & Shao, S. (2014). Spotting Fake Reviews using Positive-Unlabeled Learning. *Computación y Sistemas*, 18.
- Li, H., Meng, F., & Pan, B. (2020). How does review disconfirmation influence customer online review behavior? A mixed-method investigation. *International Journal of Contemporary Hospitality Management*, 32(11), 3685–3703. <https://doi.org/10.1108/IJCHM-03-2020-0234>
- Li, L., Ren, W., Qin, B., & Liu, T. (2015). Learning Document Representation for Deceptive Opinion Spam Detection. In M. Sun, Z. Liu, M. Zhang, & Y. Liu (Eds.), *Chinese Computational Linguistics and Natural Language Processing Based on Naturally Annotated Big Data* (pp. 393–404). Springer International Publishing. https://doi.org/10.1007/978-3-319-25816-4_32
- Lim, E.-P., Nguyen, V.-A., Jindal, N., Liu, B., & Lauw, H. W. (2010). Detecting product review spammers using rating behaviors. *Proceedings of the 19th ACM International Conference on Information and Knowledge Management*, 939–948. <https://doi.org/10.1145/1871437.1871557>
- Lim, W. M., Yap, S.-F., & Makkar, M. (2021). Home sharing in marketing and tourism at a tipping point: What do we know, how do we know, and where should we be heading? *Journal of Business Research*, 122, 534–566.
- Lin, Y., Zhu, T., Wang, X., Zhang, J., & Zhou, A. (2014). Towards online review spam detection. *Proceedings of the 23rd International Conference on World Wide Web*, 341–342. <https://doi.org/10.1145/2567948.2577293>
- Lisé, C., Larivière, V., & Archambault, É. (2008). Conference proceedings as a source of scientific information: A bibliometric analysis. *Journal of the American Society for Information Science and Technology*, 59(11), 1776–1784. <https://doi.org/10.1002/asi.20888>
- Liu, W., Jing, W., & Li, Y. (2020). Incorporating feature representation into BiLSTM for deceptive review detection. *Computing*, 102(3), 701–715. <https://doi.org/10.1007/s00607-019-00763-y>
- Lu, Y., Zhang, L., Xiao, Y., & Li, Y. (2013). Simultaneously detecting fake reviews and review spammers using factor graph model. *Proceedings of the 5th Annual ACM Web Science Conference*, 225–233. <https://doi.org/10.1145/2464464.2464470>
- Luca, M., & Zervas, G. (2016). Fake it till you make it: Reputation, competition, and Yelp review fraud. *Management Science*, 62(12), 3412–3427.
- Lundberg, S., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *ArXiv:1705.07874 [Cs, Stat]*. <http://arxiv.org/abs/1705.07874>
- Merigó, J. M., Mas-Tur, A., Roig-Tierno, N., & Ribeiro-Soriano, D. (2015). A bibliometric overview of the Journal of Business Research between 1973 and 2014. *Journal of Business Research*, 68(12), 2645–2653. <https://doi.org/10.1016/j.jbusres.2015.04.006>
- Mohawesh, R., Xu, S., Tran, S. N., Ollington, R., Springer, M., Jararweh, Y., & Maqsood, S. (2021). Fake Reviews Detection: A Survey. *IEEE Access*, 9, 65771–65802. <https://doi.org/10.1109/ACCESS.2021.3075573>
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., Altman, D., Antes, G., ... Tugwell, P. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *Journal of Chinese Integrative Medicine*, 7(9), 889–896. <https://doi.org/10.3736/jcim20090918>
- Moher, D., Shamseer, L., Clarke, M., Ghersi, D., Liberati, A., Petticrew, M., Shekelle, P., Stewart, L. A., & PRISMA-P Group. (2015). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews*, 4(1), 1. <https://doi.org/10.1186/2046-4053-4-1>
- Moon, S., Kim, M.-Y., & Iacobucci, D. (2021). Content analysis of fake consumer reviews by survey-based text categorization. *International Journal of Research in Marketing*, 38(2), 343–364. <https://doi.org/10.1016/j.jresmar.2020.08.001>

- Mukherjee, A., Liu, B., & Glance, N. (2012). Spotting fake reviewer groups in consumer reviews. *Proceedings of the 21st International Conference on World Wide Web*, 191–200. <https://doi.org/10.1145/2187836.2187863>.
- Mukherjee, S., Dutta, S., & Weikum, G. (2016). Credible Review Detection with Limited Information Using Consistency Features. In P. Frasconi, N. Landwehr, G. Manco, & J. Vreeken (Eds.), *Machine Learning and Knowledge Discovery in Databases* (pp. 195–213). Springer International Publishing. https://doi.org/10.1007/978-3-319-46227-1_13.
- Nanda, A. P., & Banerjee, R. (2021). Consumer's subjective financial well-being: A systematic review and research agenda. *International Journal of Consumer Studies*, 45(4), 750–776. <https://doi.org/10.1111/ijcs.12668>.
- Ott, M., Cardie, C., & Hancock, J. T. (2013). Negative deceptive opinion spam. *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 497–501).
- Ott, M., Choi, Y., Cardie, C., & Hancock, J. T. (2011a). Finding Deceptive Opinion Spam by Any Stretch of the Imagination. *ArXiv:1107.4557 [Cs]*. <http://arxiv.org/abs/1107.4557>.
- Ott, M., Choi, Y., Cardie, C., & Hancock, J. T. (2011b). *Finding Deceptive Opinion Spam by Any Stretch of the Imagination* (arXiv:1107.4557). arXiv. <https://doi.org/10.48550/arXiv.1107.4557>.
- Paul, J., Lim, W. M., O'Cass, A., Hao, A. W., & Bresciani, S. (2021). Scientific procedures and rationales for systematic literature reviews (SPAR-4-SLR). *International Journal of Consumer Studies*, 45(4), O1–O16. <https://doi.org/10.1111/ijcs.12695>.
- Paul, J., Merchant, A., Dwivedi, Y. K., & Rose, G. (2021). Writing an impactful review article: What do we know and what do we need to know? *Journal of Business Research*, 133, 337–340. <https://doi.org/10.1016/j.jbusres.2021.05.005>.
- Paul, J., & Singh, G. (2017). The 45 years of foreign direct investment research: Approaches, advances and analytical areas. *The World Economy*, 40(11), 2512–2527.
- Petrescu, M., O'Leary, K., Goldering, D., & Mrad, S. B. (2018). Incentivized reviews: Promising the moon for a few stars. *Journal of Retailing and Consumer Services*, 41, 288–295.
- Post, M., & Bergsma, S. (2013). Explicit and implicit syntactic features for text classification. *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 866–872.
- Rayana, S., & Akoglu, L. (2015). Collective Opinion Spam Detection: Bridging Review Networks and Metadata. *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 985–994. <https://doi.org/10.1145/2783258.2783370>.
- Ren, Y., & Ji, D. (2017). Neural networks for deceptive opinion spam detection: An empirical study. *Information Sciences*, 385–386, 213–224. <https://doi.org/10.1016/j.ins.2017.01.015>.
- Ribeiro, A. H., Tiels, K., Aguirre, L. A., & Schön, T. (2020). Beyond exploding and vanishing gradients: Analysing RNN training using attractors and smoothness. *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, 2370–2380. <https://proceedings.mlr.press/v108/ribeiro20a.html>.
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *ArXiv:1602.04938 [Cs, Stat]*. <http://arxiv.org/abs/1602.04938>.
- Roobini, M. S., Chowdary, B. N., Chowdary, J. M., Aruna, J., & Ponraj, A. (2020). Detection of fake online reviews using semi-supervised and supervised learning. *Journal of Computational and Theoretical Nanoscience*, 17(8), 3577–3580. <https://doi.org/10.1166/jctn.2020.9234>.
- Rosado-Serrano, A., Paul, J., & Dikova, D. (2018). International franchising: A literature review and research agenda. *Journal of Business Research*, 85, 238–257. <https://doi.org/10.1016/j.jbusres.2017.12.049>.
- Rout, J. K., Dalmia, A., Choo, K.-K.-R., Bakshi, S., & Jena, S. K. (2017). Revisiting semi-supervised learning for online deceptive review detection. *IEEE Access*, 5, 1319–1327. <https://doi.org/10.1109/ACCESS.2017.2655032>.
- Ruggeri, G., Orsi, L., & Corsi, S. (2019). A bibliometric analysis of the scientific literature on Fairtrade labelling. *International Journal of Consumer Studies*, 43(2), 134–152.
- Sak, H., Senior, A., & Beaufays, F. (2014). Long Short-Term Memory Based Recurrent Neural Network Architectures for Large Vocabulary Speech Recognition. *ArXiv: 1402.1128 [Cs, Stat]*. <http://arxiv.org/abs/1402.1128>.
- Salminen, J., Kandpal, C., Kamel, A. M., Jung, S., & Jansen, B. J. (2022). Creating and detecting fake reviews of online products. *Journal of Retailing and Consumer Services*, 64, Article 102771. <https://doi.org/10.1016/j.jretconser.2021.102771>.
- Satia Budhi, G., Chiong, R., Wang, Z., & Dhakal, S. (2021). Using a hybrid content-based and behaviour-based featuring approach in a parallel environment to detect fake reviews. *Electronic Commerce Research and Applications*, 47, Article 101048. <https://doi.org/10.1016/j.elerap.2021.101048>.
- Saumya, S., & Singh, J. P. (2022). Spam review detection using LSTM autoencoder: An unsupervised approach. *Electronic Commerce Research*, 22(1), 113–133. <https://doi.org/10.1007/s10660-020-09413-4>.
- Savage, D., Zhang, X., Yu, X., Chou, P., & Wang, Q. (2015). Detection of opinion spam based on anomalous rating deviation. *Expert Systems with Applications*, 42(22), 8650–8657. <https://doi.org/10.1016/j.eswa.2015.07.019>.
- Shafer, G. (1976). *A Mathematical Theory of Evidence*. Princeton University Press.
- Shah, N., Beutel, A., Hooi, B., Akoglu, L., Gunnemann, S., Makhija, D., Kumar, M., & Faloutsos, C. (2016). EdgeCentric: Anomaly Detection in Edge-Attributed Networks. *2016 IEEE 16th International Conference on Data Mining Workshops (ICDMW)*, 327–334. <https://doi.org/10.1109/ICDMW.2016.0053>.
- Sheibani, A. A. (2012). Opinion mining and opinion spam: A literature review focusing on product reviews. *6th International Symposium on Telecommunications (IST)*. <https://doi.org/10.1109/ISTEL.2012.6483152>.
- Small, H. (1973). Co-citation in the scientific literature: A new measure of the relationship between two documents. *Journal of the American Society for Information Science*, 24(4), 265–269.
- Smets, P., & Kennes, R. (2008). The Transferable Belief Model. In R. R. Yager, & L. Liu (Eds.), *Classic Works of the Dempster-Shafer Theory of Belief Functions* (pp. 693–736). Springer. https://doi.org/10.1007/978-3-540-44792-4_28.
- Socher, R., Chen, D., Manning, C. D., & Ng, A. (2013). Reasoning with neural tensor networks for knowledge base completion. *Advances in Neural Information Processing Systems*, 926–934.
- Socher, R., Pennington, J., Huang, E. H., Ng, A. Y., & Manning, C. D. (2011). Semi-supervised recursive autoencoders for predicting sentiment distributions. *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*, 151–161.
- Södergren, J. (2021). Brand authenticity: 25 Years of research. *International Journal of Consumer Studies*, 45(4), 645–663. <https://doi.org/10.1111/ijcs.12651>.
- Souppouris, A. (2013, October 24). *Samsung fined \$340,000 for faking online comments*. The Verge. <https://www.theverge.com/2013/10/24/5023658/samsung-fined-340000-for-posting-negative-htc-reviews>.
- Sun, H., Morales, A., & Yan, X. (2013). Synthetic review spamming and defense. *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1088–1096. <https://doi.org/10.1145/2487575.2487688>.
- Usai, A., Pironti, M., Mital, M., & Mejri, C. A. (2018). Knowledge discovery out of text data: A systematic review via text mining. *Journal of Knowledge Management*.
- Vidanagama, D. U., Silva, T. P., & Karunananda, A. S. (2020). Deceptive consumer review detection: A survey. *Artificial Intelligence Review*, 53(2), 1323–1352. <https://doi.org/10.1007/s10462-019-09697-5>.
- Wang, G., Xie, S., Liu, B., & Yu, P. (2012). Identify online store review spammers via social review graph. *ACM Transactions on Intelligent Systems and Technology*, 1–21.
- Wang, S. I., & Manning, C. D. (2012). Baselines and bigrams: Simple, good sentiment and topic classification. *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 90–94.
- Wang, W., Li, F., & Yi, Z. (2019). Scores vs. stars: A regression discontinuity study of online consumer reviews. *Information & Management*, 56(3), 418–428. <https://doi.org/10.1016/j.im.2018.08.002>.
- Wang, Z., Gu, S., Zhao, X., & Xu, X. (2018). Graph-based review spammer group detection. *Knowledge and Information Systems*, 55(3), 571–597. <https://doi.org/10.1007/s10115-017-1068-7>.
- Wu, Y., Ngai, E. W. T., Wu, P., & Wu, C. (2020). Fake online reviews: Literature review, synthesis, and directions for future research. *Decision Support Systems*, 132, Article 113280. <https://doi.org/10.1016/j.dss.2020.113280>.
- Xie, S., Wang, G., Lin, S., & Yu, P. S. (2012). Review spam detection via temporal pattern discovery. *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 823–831. <https://doi.org/10.1145/2339530.2339662>.
- Xu, C., Zhang, J., Chang, K., & Long, C. (2013). Uncovering Collusive Spammers in Chinese Review Websites. *Proceedings of the 22nd ACM International Conference on Information & Knowledge Management*, 979–988. <https://doi.org/10.1145/2505515.2505700>.
- Xu, Q., & Zhao, H. (2012). Using Deep Linguistic Features for Finding Deceptive Opinion Spam. *Proceedings of COLING 2012: Posters*, 1341–1350. <https://aclanthology.org/C12-2131>.
- Xu, Z., Ge, Z., Wang, X., & Skare, M. (2021). Bibliometric analysis of technology adoption literature published from 1997 to 2020. *Technological Forecasting and Social Change*, 170, Article 120896. <https://doi.org/10.1016/j.techfore.2021.120896>.
- Xue, H., Li, F., Seo, H., & Pluretti, R. (2015). Trust-Aware Review Spam Detection. *2015 IEEE Trustcom/BigDataSE/ISPA*, 1, 726–733. <https://doi.org/10.1109/Trustcom.2015.440>.
- Yao, Y., Viswanath, B., Cryan, J., Zheng, H., & Zhao, B. Y. (2017). *Automated crowdurfing attacks and defenses in online review systems*. 1143–1158.
- Yu, C., Zuo, Y., Feng, B., An, L., & Chen, B. (2019). An individual-group-merchant relation model for identifying fake online reviews: An empirical study on a Chinese e-commerce platform. *Information Technology and Management*, 20(3), 123–138. <https://doi.org/10.1007/s10799-018-0288-1>.
- Zhang, D., Zhou, L., Kehoe, J. L., & Kilic, I. Y. (2016). What online reviewer behaviors really matter? Effects of verbal and nonverbal behaviors on detection of fake online reviews. *Journal of Management Information Systems*, 33(2), 456–481. <https://doi.org/10.1080/07421222.2016.1205907>.
- Zhang, W., Du, Y., Yoshida, T., & Wang, Q. (2018). DRI-RCNN: An approach to deceptive review identification using recurrent convolutional neural network. *Information Processing & Management*, 54(4), 576–592. <https://doi.org/10.1016/j.ipm.2018.03.007>.
- Zhao, S., Xu, Z., Liu, L., Guo, M., & Yun, J. (2018). Towards Accurate Deceptive Opinions Detection Based on Word Order-Preserving CNN. *Mathematical Problems in Engineering*, 2018, e2410206. <https://doi.org/10.1155/2018/2410206>.

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